

26. What is Recursion?

Answer: Recursion is when a function calls itself until a condition is met.

Real-World Example: Climbing stairs one step at a time until you reach the top.

27. What is Exception Handling?

Answer: Exception handling is used to handle errors without stopping the program.

Real-World Example: Wearing a helmet while riding a bike for safety during unexpected situations.

28. What is a Module?

Answer: A module is a Python file that contains reusable code such as functions and classes.

Real-World Example: A toolbox that contains different tools for different jobs.

29. What is the difference between `append()` and `extend()`?

Answer: `append()` adds one item to a list, while `extend()` adds multiple items from another list.

Real-World Example: Adding one book to your bag (`append`) vs adding all the books from another bag (`extend`).

30. Why should we learn Python?

Answer: We should learn Python because it is easy to learn and widely used in software development, automation, AI, and data science.

Real-World Example: Learning English helps you communicate in many countries; similarly, learning Python helps you build many types of applications.